COMPUTER SCIENCE & DA

Data Structure through Python

Trees

DPP- 01

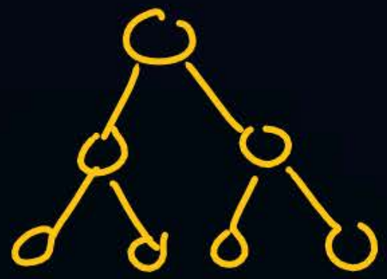
Discussion Notes



Mohd Kashif Khan sir

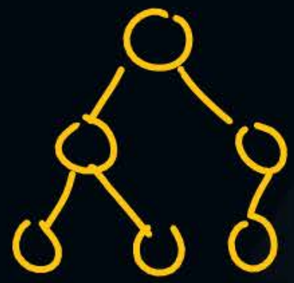
[MCQ]

#Q Which of the below statement is/are Invalid?



A. Every Perfect Binary Tree is a Complete Binary Tree

B. Every Complete Binary Tree is a Full Binary Tree



C. Every Full Binary Tree is a Complete Binary Tree

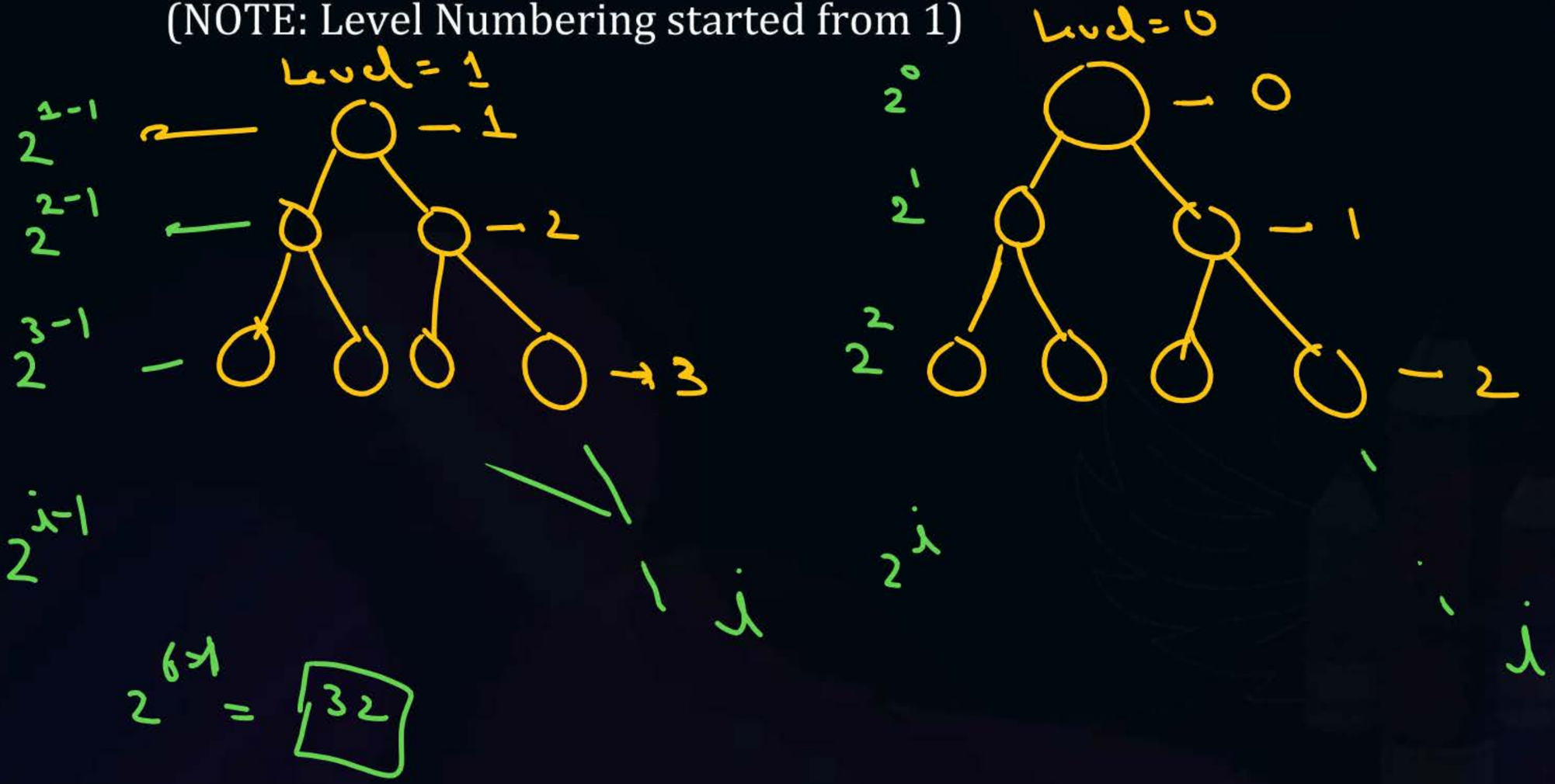


D. Every Full Binary Tree is a Perfect Binary Tree

[NAT]

#Q The Number Of Nodes in a Perfect binary tree at level 6 will be 32

(NOTE: Level Numbering started from 1)



[NAT]

#Q The number of leaf nodes in a binary tree, if there are 6 nodes with 2 children is _____.

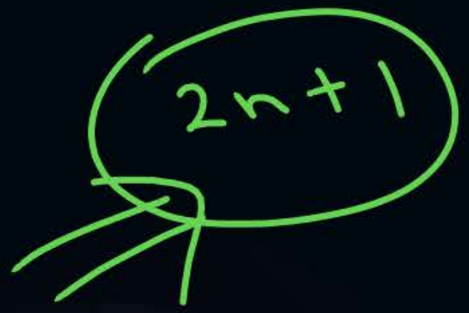
7

n nodes with 2 children

leaf nodes = n + 1

[MCQ]

#Q The Minimum number of nodes with height 2n in a binary tree will be _____

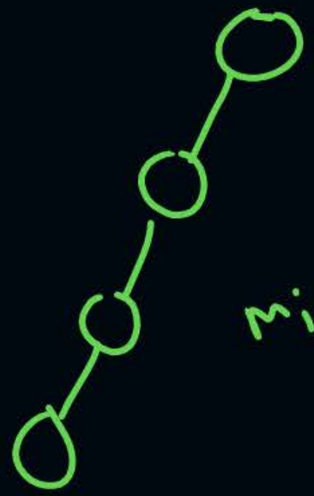


A n

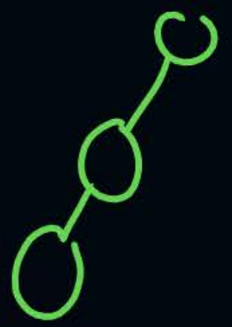
B n + 1

C n – 1

D 2n



Height = 3
Min. No of nodes = 4



Height = 2
Min no. of nodes = 3

[MCQ]

#Q The number Of labelled binary trees with 4 nodes is ____

A 14

B 24

C 336

D 70

$$\frac{{}^{2n}C_n}{n+1} \times n!$$

$$\frac{{}^{8}C_4}{5} \times 4! = \frac{8!}{4!\,4!\,5} \times 4!$$

$$= \frac{8 \times 7 \times 6 \times 5 \times 4!}{4! \times 5}$$

$$= 56 \times 6$$

$$= 336$$

[MCQ]

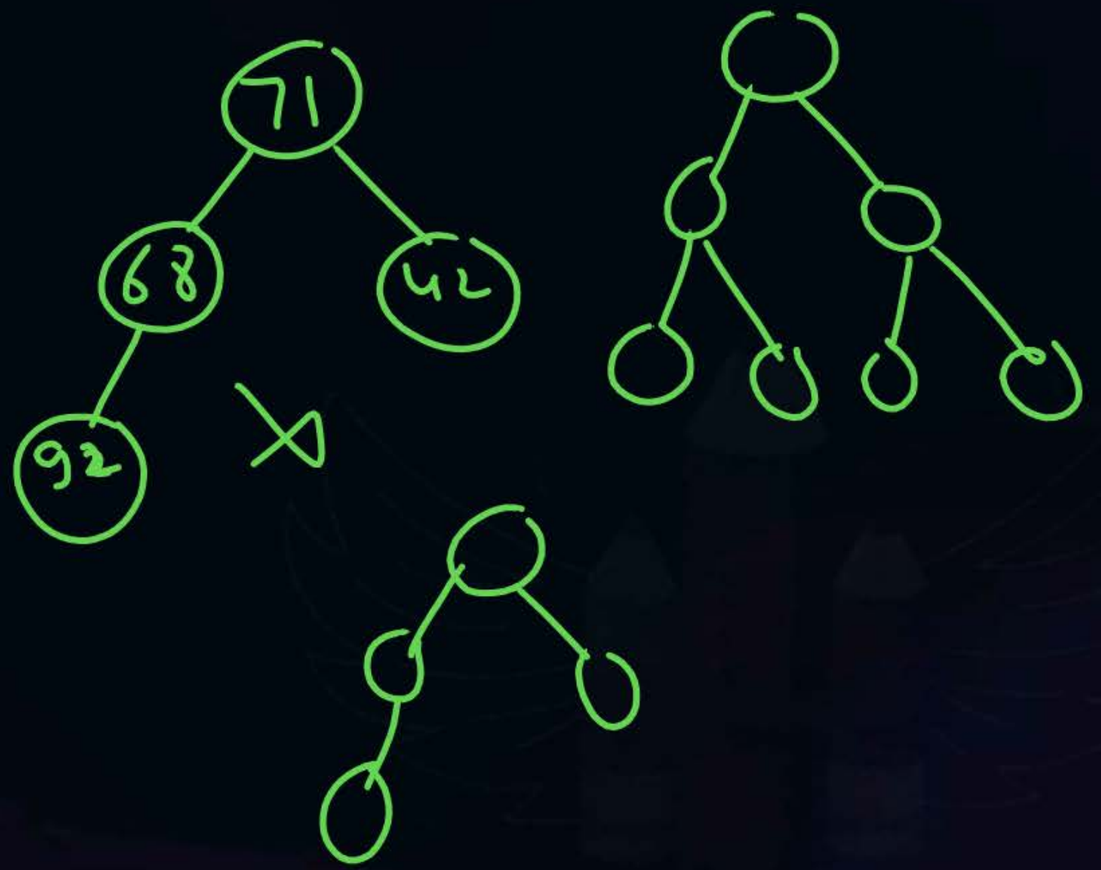
#Q Identify False statement(s) from below:

A Every Binary Heap is a CBT.

B Every CBT is a Binary Heap.

C Every PBT is a CBT.

D Every CBT is a PBT



[MCQ]

#Q The number of Unlabelled and Labelled binary trees possible with 5 elements is _____ and _____, respectively.

A 7, 840

B 42, 840

C 7, 5040

D 42, 5040

$\frac{2nC_n}{n+1}$

$\frac{2nC_n}{n+1} \times n!$

$\frac{10C_5}{6} = \frac{10!}{5! \times 5! \times 6}$

$= \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5!}{5! \times 5! \times 6}$

$= \frac{10 \times 9^3 \times 8^2 \times 7}{8 \times 4 \times 3 \times 4} = 42$

42×120

$= 5040$

[MCQ]

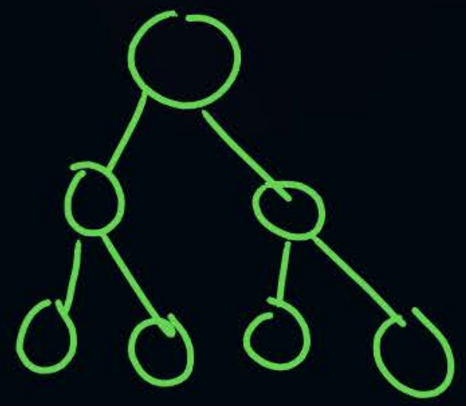
#Q In _____ binary tree, the number of nodes will be maximum with minimum height.

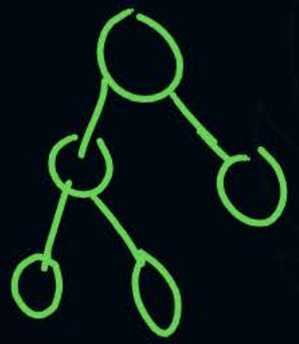
A Full binary Tree

B Skewed binary tree

C Perfect Binary Tree

D Degenerated binary Tree







[MCQ]

#Q Consider a full binary tree with 15 leaf nodes.
Then, the number of internal nodes and total nodes in tree are respectively ___14___ and ___29___.

A 15, 30

B 14, 29

C 15, 31

D 16, 31

no. of leaf nodes = 15
No of internal nodes = 15 - 1 = 14
Total no. of nodes = 15 + 14 = 29

[MCQ]

#Q The minimum height of binary tree possible with 12 Nodes is _____.

A 2

B 3

C 4

D 5

$$\lceil \log_2^{h+1} \rceil - 1$$

$$= \lceil \log_2^{12+1} \rceil - 1$$

$$= \lceil \log_2^{13} \rceil - 1$$

$$= 4 - 1 = \boxed{3}$$

THANK - YOU